HM3300

500 mA, 3.3 V fixed low-dropout (LDO) linear regulator

Features

- Input voltage range: 2.5 V to 6.0 V
- Fixed 3.3 V output, ±1% accuracy at 25 °C
- Output current up to 500 mA
- Dropout voltage: 180 mV typical at 500 mA
- Quiescent current: 45 µA typical
- PSRR: 70 dB at 1 kHz
- Thermal shutdown at 160 °C with 20 °C hysteresis
- Active-high enable input
- Available in SOT-23-5 and DFN-6 packages

General description

The HM3300 is a low-noise, low-quiescent-current LDO regulator intended for battery-powered and noise-sensitive designs. It is stable with a 1 µF ceramic output capacitor and includes current limit, thermal shutdown and a shutdown mode that reduces supply current to 0.1 µA typical.

Applications

Wearables, Sensor nodes, BLE modules, Portable instruments.

Ordering information

Order code	Package	Packing	Temperature grade
HM3300-SOT5-T	SOT-23-5	Tape & reel, 3000	–40 °C to 125 °C
HM3300-DFN6-T	DFN-6 (2×2 mm)	Tape & reel, 5000	–40 °C to 125 °C

Absolute maximum ratings

Stresses beyond these ratings may cause permanent damage. Functional operation is not implied.

Parameter	Min	Max	Unit
VIN to GND	–0.3	7.0	V
EN to GND	–0.3	VIN + 0.3	V
Output short-circuit duration	—	Continuous	
ESD, human body model	—	2	kV
Junction temperature TJ	—	150	°C
Storage temperature	–65	150	°C

Recommended operating conditions

Parameter	Min	Typ	Max	Unit
Input voltage VIN	2.5	—	6.0	V
Output current IOUT	0	—	500	mA
Ambient temperature TA	–40	—	125	°C
Input capacitor CIN	1	—	—	µF
Output capacitor COUT (ceramic)	1	—	—	µF

Electrical characteristics

Typical values at TA = 25 °C unless otherwise noted.

Parameter	Conditions	Min	Typ	Max	Unit
Output voltage	IOUT = 1 mA, VIN = 4.3 V	3.267	3.300	3.333	V
Line regulation	VIN = 3.8 V to 6.0 V	—	0.02	0.1	%/V
Dropout voltage	IOUT = 500 mA	—	180	300	mV
Quiescent current	IOUT = 0 mA	—	45	70	µA
Shutdown current	EN = 0 V	—	0.1	1	µA
Current limit	VOUT = 0 V	520	650	900	mA
PSRR	f = 1 kHz, IOUT = 100 mA	—	70	—	dB
Output noise	10 Hz to 100 kHz	—	38	—	µVrms
Thermal shutdown threshold		—	160	—	°C
EN input high threshold		1.2	—	—	V
EN input low threshold		—	—	0.4	V

Pin configuration and functions

Pin	Name	Type	Description
1	VIN	Power	Input supply. Bypass with at least 1 µF ceramic capacitor close to the pin.
2	GND	Ground	Ground reference.
3	EN	Input	Enable. Drive high (≥ 1.2 V) to enable, low (≤ 0.4 V) to shut down.
4	NC	—	No connection. Leave floating or connect to GND.
5	VOUT	Power	Regulated 3.3 V output. Bypass with at least 1 µF ceramic capacitor.

Package and thermal information

Package	Thermal resistance θJA	Dimensions
SOT-23-5	θJA = 220 °C/W	2.9 × 1.6 mm
DFN-6 (2×2 mm)	θJA = 65 °C/W	2.0 × 2.0 mm

Application information

Power dissipation is PD = (VIN − VOUT) × IOUT. Keep the junction temperature below 125 °C for rated operation; see application note AN-102 for worked examples.

For lowest noise, place the 1 µF output capacitor within 3 mm of VOUT and ground it directly to the GND pin.

Revision history

Revision	Date	Change
Rev A	2024-03-02	Initial release
Rev B	2024-09-18	Added DFN-6 package; updated PSRR
Rev C	2025-04-07	Corrected dropout maximum to 300 mV